TIPA Korea Technology and Information Promotion Agency for SMEs

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Digital mobility biomarkers: a technology that allows the early detection and prevention of functional decline in old age, neurological disorders, and mobility-limiting diseases.**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio・Medical		Bio・Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Information & Systems	Remote and Home Health care Devices	Therapeutic Devices & Diagnostic Devices	Bio-signal Measurement / Diagnostic Devices
Concept of Proposed Technology	CSEM develops technologies and algorithms for analyzing and tracking motion across multiple body locations, including the wrist, foot, thorax, and head. Our algorithm portfolio covers activity classification, gait analysis, fall detection, workout profiling, sleep monitoring, and motion and gesture recognition. Our step detector has been independently verified as best-in-class for accuracy. All solutions are designed for real-time operation with minimal power and memory requirements, making them ideal for embedded Digital Health and fitness applications. Specific application of gait analysis includes mobility analysis (gait speed, symmetry, balance), neurological disease assessment (Parkinson's disease, Alzheimer's), real-time feedback (e.g. for diabetes management, load monitoring after surgery, gait retraining). Going beyond the state of the art, the proposed technology would move from standard gait parameters which may lack predictive/preventive information to digital biomarkers linked to disease progression and health status. These biomarkers can be closely tailored to each user and provide them with understandable added value that today's standard technologies do not achieve.			

Rationale for Joint R&D Project	The gap between technology and product realization remains wide. While CSEM's gait analysis solutions are highly accurate, they should be demonstrated in different populations and for various clinical applications. The discovery of digital biomarkers and their application in digital health is an emerging field poised to support healthcare through prevention and improved disease management. Therefore, joint R&D projects are important and should lead to new products/applications that cater to patients in the Digital Health space.
Key Words	1. Digital Biomarkers 2. Gait Analysis 3. Artificial Intelligence 4. Mobility Assessment 5. Inertial Measurement Units

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 7	End of the Project	Level 9
R&D Objectives	The main R&D objective is to discover digital mobility biomarkers that are relevant to the management of neurological disorders, aging, and mobility-related diseases. Going beyond our state-of-the-art gait analysis modules, we would like to build new knowledge that can be transferred to companies in this field.			
R&D Target Specifications	We target: a) Temporal gait parameter accuracy > 95% and delay <60ms b) Spatial gait parameter accuracy >90% c) Gait biomarker correlation with mobility/disease clinical metrics >0.9			
R&D Plan	Provide detailed plan outlining the research approach, key tasks, methodologies, expected milestones, and the timeline required to achieve the stated R&D objectives. The research approach involves: a) Setting up the requirements with R&D partners b) Collecting the necessary data to build digital biomarkers with specific populations where the digital health needs are still unmet. c) Developing Artificial Intelligence algorithms building on biomechanical models of human movement to derive digital mobility biomarkers d) Verifying and validating these biomarkers in clinical pilots e) Transferring the developed technologies through licensing to commercial partners in Korea CSEM can support our customers with the entire development chain: from hardware reference design to electronics component selection, firmware/software development, system integration, algorithm implementation, and cloud/edge analytics.			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	CSEM's Business Unit Medtech is home to interconnected thematic groups covering a) Medical Systems, b), c) Signal Processing and AI, c) Electronics and Sensing, d) IoMT & Cloud Software, and e) Prototyping and Firmware. The business unit counts 60 people dedicated to building the digital health world of tomorrow. CSEM can therefore function as an applied research partner for the development of digital biomarker discovery solutions and support companies at different stages of their product design. In this scope, we can play the role of algorithm development experts, support clinical/data collection trials, build data pipelines to collect sensor data remotely, review device schematics and propose optimizations, provide reference designs and firmware. We have a prototyping lab for small series manufacturing, and a physiological/sports lab to perform data collection with e.g. treadmill or ergobike and are in close contact with gait analysis laboratories in Bern with which we perform clinical studies.
Expected R&R of Korean SME Partner	The expected roles and responsibilities of Korean SME can be for example: product and system design, elaboration of requirements, product roadmap planning, verification and validation, planning data collection studies, among others. However, CSEM collaborates with customers at all levels of their product development, so we are flexible in task distribution between partners.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Explain the expected outcomes of this joint R&D project, including technological achievements, commercialization potential, and broader industrial or economic impact, etc. The expected outcomes of a joint R&D project are: a) Innovative digital biomarkers that relate to prevention, disease management, clinical intervention tailoring, and decision support systems. b) New products/digital solutions for health and mobility monitoring, providing novel features to support longevity and better psychophysiological health. c) Novel hardware/software and cloud/edge ecosystems that seamlessly integrate into users' daily lives without stigmatization and excessive user interaction. d) Healthcare cost reduction through remote monitoring and feedback

6. References

(1)

(2)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization